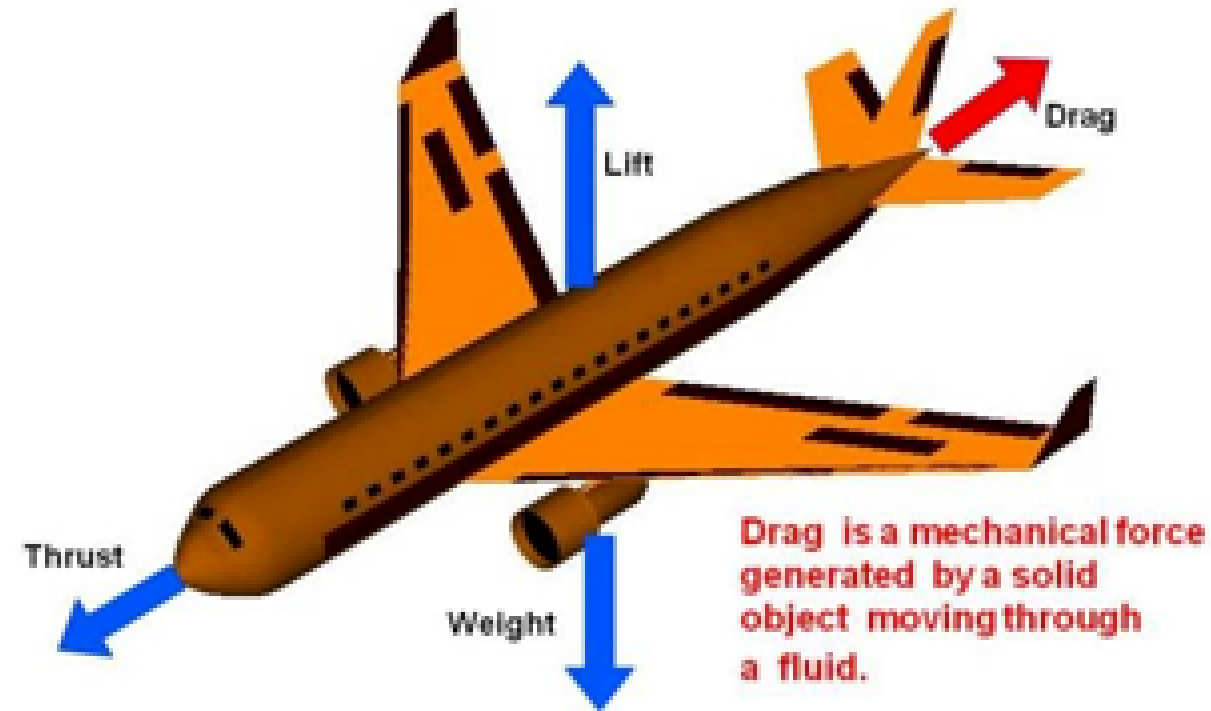


Day 07 - Drag Forces



Announcements

- Homework 2 is due Friday
- Video recordings have continued to fail.
 - Zoom password: phy321
- Updated office hours (Danny-DC; Elisha-EA):
 - Monday 4-5pm (DC) - change?
 - Tuesday 5-6pm (EA)
 - Wednesday 4-5pm (DC)
 - Thursday 5-6pm (EA)
 - Friday 10-12pm (DC then EA); 3-4pm (DC)

Seminars this week

WEDNESDAY, January 29, 2025

- Astronomy Seminar, 1:30 pm, 1400 BPS, Michiel Lambrechts, Univ. of Copenhagen, *Planet formation*
- FRIB Nuclear Science Seminar, 3:30pm., FRIB 1300 Auditorium, Brenden Longfellow of Lawrence Livermore National Laboratory, *From Tensor Current Limits to Solar Neutrinos: ^8Li and ^8B Studies with the Beta-decay Paul Trap*

Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Clicker Question 6-2

Assuming a **linear model** for Air Resistance $\sim bv$, we obtained this EOM for a falling ball:

$$\ddot{y} = -g + \frac{b}{m}\dot{y}$$

What happens when $\ddot{y} = 0$?

1. The ball stops moving ($v = 0$).
2. The ball reaches a velocity of mg/b .
3. The ball reaches a terminal velocity.
4. I'm not sure.

Clicker Question 6-3

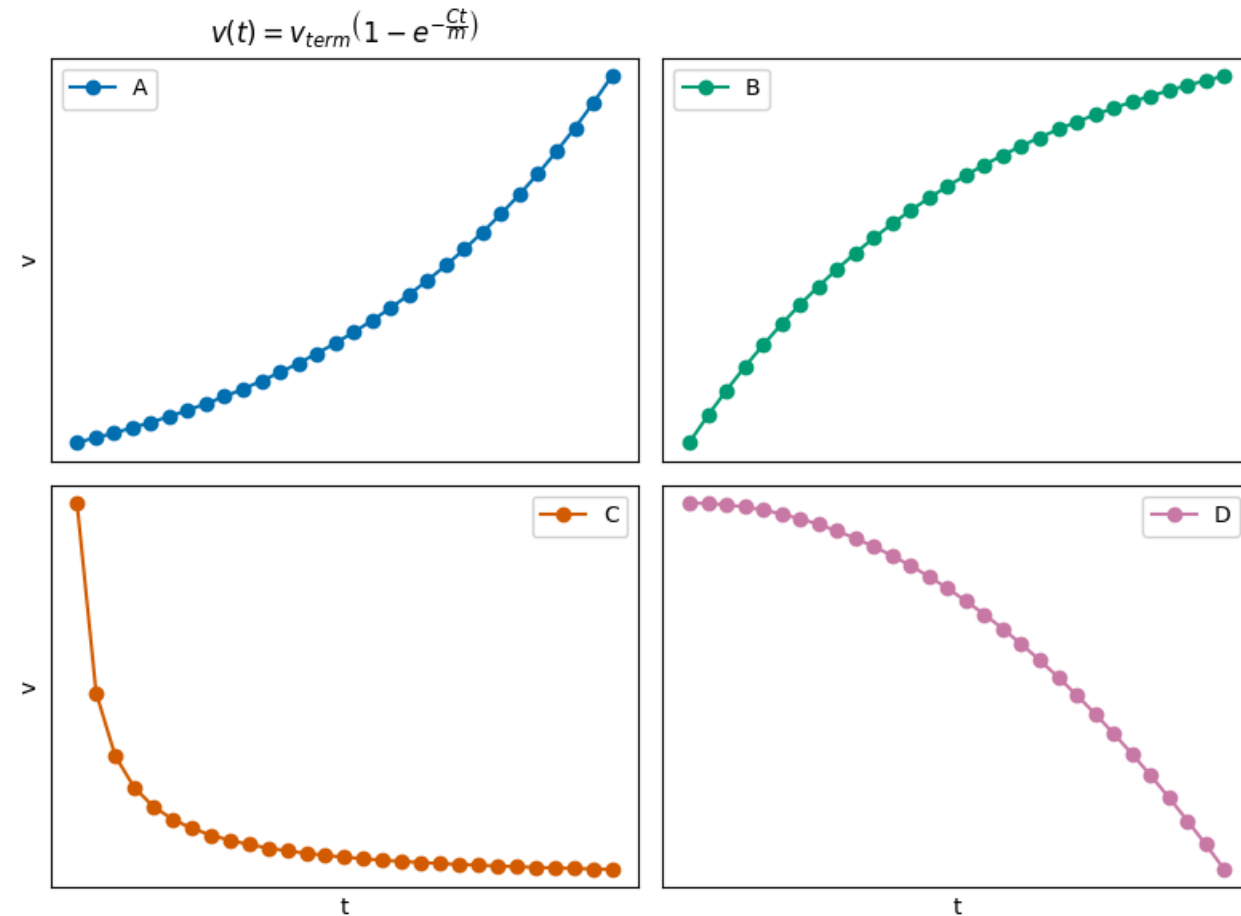
For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units of $[gt/v_{term}]$?

1. Yes, the units for $[gt/v_{term}]$ are m/s ; both sides have the same units.
2. No, the units for $[gt/v_{term}]$ are m/s ; each side has different units.
3. Yes, the units for $[gt/v_{term}]$ are unit-less; both sides have the same units.
4. No, the units for $[gt/v_{term}]$ are unit-less; each side has the different units.

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.